

Kshitij Patil

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[github](#) | [linkedin](#) | [Portfolio](#)

EDUCATION

Master of Science - Computer Science

University at Buffalo, The State University of New York

Aug 2022 - Dec 2023

Bachelor of Engineering - Computer Science

Smt. Kashibai Navale College of Engineering

Aug 2016 - Jun 2020

EXPERIENCE

TikTok

Feb 2025 - Present

Site Reliability Engineer

San Jose, CA

- Led **osquery** deployment across **170K+** production machines, sustaining **99.9%** agent uptime and fleet visibility.
- Built a **Kafka**-based **telemetry pipeline** handling **2M+** events/min with sub-second delivery to **SIEM**.
- Operated **ZTI** control plane on **Kubernetes**, backed by **SPIFFE/SPIRE**, supporting **170M+** users.
- Spearheaded **ArgoCD**-driven **GitOps** workflows in deployment pipelines, optimizing efficiency by **50%**.

Gala Games

Oct 2024 - Feb 2025

Site Reliability Engineer

San Francisco, CA

- Leveraged **Kubernetes Operators** and **CRDs** to manage a Layer 1 blockchain hosting **\$600M** in NFTs.
- Cut compute costs by **25%** by shifting **EKS** workloads to **Karpenter**-provisioned **Spot** and tuning **autoscaling**.
- Integrated **Prometheus** and **Grafana** for real-time monitoring, improving system observability by **35%**.
- Secured **\$600M** in cryptocurrency assets by backing up **production Kubernetes clusters** with **Velero**.

Incalus Inc

May 2023 - Oct 2024

Site Reliability Engineer

Fremont, CA

- Configured a **Kubernetes** cluster from scratch with **kubeadm** (control plane + workers), cutting costs by **34%**.
- Engineered **CI/CD pipelines** with **GitLab CI/CD** and **GitHub Actions**, accelerating deployment cycles by **46%**.
- Implemented **NGINX Ingress Controller** in **Kubernetes**, refining routing and enhancing app scalability by **20%**.

Mckinley Rice

Dec 2021 - Apr 2022

Site Reliability Engineer

Pune, IN

- Automated **infrastructure provisioning** with **Terraform**, streamlining setup and reducing time by **43%**.
- Reduced **MTTR** by **63%** through **log monitoring** and **Python automation** scripts to prevent outages.

Cognizant

Nov 2020 - Dec 2021

Programmer Analyst - Site Reliability

Pune, IN

- Reduced **microservices** deployment time by **45%** with **Github Actions**, **SonarQube**, **AWS ECR**, **ECS**, and **Docker**.
- Built **Kibana** and **Splunk** dashboards for **service latency** and **error rate**, reducing detection time by **70%**.

PROJECTS

Infrastructure Management with Terraform [GitHub](#)

- Implemented **Terraform** for **AWS infrastructure**, including **EC2**, **DynamoDB**, **S3 backend** and **Workspaces**.

End-to-End CI Pipeline with SonarQube Integration [GitHub](#)

- Architected a **Jenkins** pipeline with stages for **testing**, **coverage**, **SonarQube** analysis, and **Docker** deployment.

Blockchain Infrastructure Deployment using Kubernetes [GitHub](#)

- Deployed a **Blockchain Network** on **Kubernetes**, using **Deployments**, **Jobs**, **ConfigMaps**, **Secrets**, and **PVs**.

SKILLS

Languages: Go, Python, Bash, Shell Scripting, Java, Node.js, JavaScript, TypeScript, Ruby, PowerShell

Cloud: AWS-EC2, S3, RDS, NLB, ALB, Auto Scaling, IAM, VPC, Route 53, Lambda, Azure, GCP

Containerization: Kubernetes, Istio, Service Mesh, EKS, ECS, ECR, Docker, Helm

Infrastructure as code: Terraform, Ansible, CloudFormation, CDK, Puppet, Chef, Salt

CI/CD: Jenkins, GitHub Actions, GitLab CI, Git, CircleCI, ArgoCD, FluxCD

Database Management: MySQL, PostgreSQL, MongoDB, DynamoDB, Elasticsearch, Redis, Kafka

Methodologies and OS: Agile, JIRA, Disaster Recovery, GraphQL, TCP/IP, Troubleshooting, Linux, Windows

Monitoring and Logging: Prometheus, Grafana, ELK Stack, CloudWatch, Datadog, Splunk, Kibana

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)

Badge

Certified Kubernetes Application Developer (CKAD)

Badge